Vegetarian diets and blood pressure: a meta-analysis.

IMPORTANCE Previous studies have suggested an association between vegetarian diets and lower blood pressure (BP), but the relationship is not well established. OBJECTIVE To conduct a systematic review and meta-analysis of controlled clinical trials and observational studies that have examined the association between vegetarian diets and BP. DATA SOURCES MEDLINE and Web of Science were searched for articles published in English from 1946 to October 2013 and from 1900 to November 2013, respectively. STUDY SELECTION All studies met the inclusion criteria of the use of (1) participants older than 20 years, (2) vegetarian diets as an exposure or intervention, (3) mean difference in BP as an outcome, and (4) a controlled trial or observational study design. In addition, none met the exclusion criteria of (1) use of twin participants, (2) use of multiple interventions, (3) reporting only categorical BP data, or (4) reliance on case series or case reports. DATA EXTRACTION AND SYNTHESIS Data collected included study design, baseline characteristics of the study population, dietary data, and outcomes. The data were pooled using a random-effects model. MAIN OUTCOMES AND MEASURES Net differences in systolic and diastolic BP associated with the consumption of vegetarian diets were assessed. RESULTS Of the 258 studies identified, 7 clinical trials and 32 observational studies met the inclusion criteria. In the 7 controlled trials (a total of 311 participants; mean age, 44.5 years), consumption of vegetarian diets was associated with a reduction in mean systolic BP (-4.8 mm Hg; 95% CI, -6.6 to -3.1; $P < .001$; $I2 = 0$; $P = .45$ for heterogeneity) and diastolic BP (-2.2 mm Hg; 95% CI, -3.5 to -1.0; $P < .001$; $I2 = 0$; $P = .43$ for heterogeneity) compared with the consumption of omnivorous diets. In the 32 observational studies (a total of 21,604 participants; mean age, 46.6 years), consumption of vegetarian diets was associated with lower mean systolic BP (-6.9 mm Hg; 95% CI, -9.1 to -4.7; $P < .001$; $I2 = 91.4$; $P < .001$ for heterogeneity) and diastolic BP (-4.7 mm Hg; 95% CI, -6.3 to -3.1; $P < .001$; $I2 = 92.6$; $P < .001$ for heterogeneity) compared with the consumption of omnivorous diets. CONCLUSIONS AND RELEVANCE Consumption of vegetarian diets is associated with lower BP. Such diets could be a useful nonpharmacologic means for reducing BP.